FR3 UAV Channel Knowledge Maps: A Dataset and an Interpretable Attenuation Model

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***Abstract*—This paper introduces a dense FR3 channel knowledge map dataset for unmanned aerial vehicle base stations. It contains 16180 ray-traced realizations from 3236 locations across 66 cities at 7.125 GHz, with attenuation, geometric visibility, delay spread, and angular spread stored at 513×513 resolution for five transmitter heights per location. We use attenuation as a focused first benchmark and propose a height-aware, auditable model. Its LoS branch combines free-space loss with a coherent image-source two-ray term, whereas its NLoS branch combines a COST231-shaped basis with local morphology and topology-specific linear calibration. The dataset-supplied ray-traced visibility mask selects the branch. Under a strict city holdout, the frozen model reaches 1.9277 dB overall root mean square error, with 1.7370 dB in LoS and 3.5275 dB in NLoS, on 2590 maps from 14 unseen cities. Its median CPU time is 0.0602 s per map. The result provides a transparent attenuation baseline and a physical anchor for later distribution-aware refinement of all three channel quantities.**

***Index Terms*—channel attenuation, channel knowledge map, delay spread, path loss, radio map, unmanned aerial vehicle.**

I. Introduction

Uncrewed aerial vehicles (UAVs) carrying base stations can support future 6G networks through mobility and an increased probability of line of sight (LoS). Their deployment nevertheless requires fast location-dependent channel information for placement and resource management. Channel knowledge maps (CKMs) address this need by indexing channel information by transmitter or receiver location [1], [2]. A CKM may store attenuation, multipath delays and angles, or richer channel descriptions.

CKMs can be obtained from measurements or channel models. Measurements are accurate but costly to collect densely, whereas deterministic ray tracing (RT) offers spatially complete labels at substantial computational cost. Available CKM datasets commonly rely on RT [3], [4], and learned surrogates such as RadioUNet and PMNet use convolutional encoder–decoders to predict dense channel maps from city geometry and transmitter information [5]–[7].

The upper mid-band frequency range FR3, between 7 and 24 GHz, is a candidate for combining wider bandwidth than FR1 with more favorable coverage than FR2 [8]. It is especially relevant to elevated transmitters, which often benefit from LoS yet must still operate through urban NLoS regions. Assessing that setting requires channel data across varied building morphologies and transmitter heights. To our knowledge, no public CKM dataset currently provides this combination in FR3.

a) Limitations of the existing CKM datasets: Table I shows the remaining gap: existing dense sets use FR1 or FR2, lack UAV transmitter heights, or do not provide building-scale height information. UAV channels can vary substantially with altitude and accurate urban blockage requires building height as well as footprint. This information is sparse in OpenStreetMap [10]. We therefore use global building-scale products that provide both quantities [11], [12].

b) Limitations of the existing CKM modeling approaches: Recent CKM estimators use increasingly expressive convolutional, transformer, variational, and diffusion architectures. Such models are valuable when sparse observations or unresolved propagation effects must be inferred, but their added complexity is rarely isolated from stable propagation structure. This comparison is particularly relevant when geometric visibility is already available and transmitter height varies continuously. A calibrated analytical base can explain the repeatable structure before a learned model is asked to represent local residuals and multimodal tails.

c) Contributions: The paper makes two contributions:

1) We introduce an FR3 CKM dataset generated by RT at 3236 locations across 66 cities. Five transmitter heights per location yield 16180 dense maps containing attenuation, LoS, delay spread, and angular spread. Global building footprint and height products provide the underlying 3D geometry [11], [12].
2) We establish an attenuation benchmark through a height-aware model that separates LoS and NLoS mechanisms. Coherent two-ray structure handles LoS, while topology-specific linear calibration corrects a COST231-shaped NLoS basis. Every fitted quantity is frozen before evaluation on unseen cities.

Section II defines the data and evaluation contract, Section III presents the attenuation model, Section IV reports the results, and Section V concludes the paper.

II. Data and Evaluation Contract

Each sample is a 513×513 urban CKM with $1\,\mathrm{m}$ spatial resolution. One elevated transmitter is horizontally centered in the map, operates at 7.125 GHz, and has a sample-dependent height $h_{\mathrm{tx}} \approx 12\,\mathrm{m}$ to $478\,\mathrm{m}$. Every non-building pixel is a receiver candidate at $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$. The HDF5 stores dense attenuation, delay spread in ns, angular spread in degrees, and geometric LoS for every realization. All three channel quantities are released; the model and benchmark in this paper

TABLE I
COMPARISON OF DENSE CKM DATASETS.

Property	This work	UrbanRadio3D [9]	RadioMapSeer [3]	CKMImageNet [4]
Frequency range	FR3	FR1	FR1	FR2
Elevated Tx height	yes	no	no	no
Number of maps	16180	701	701	42
Map size	513×513	256×256	256×256	128×128
Horizontal resolution	1 m	1 m	1 m	2 m
Real building footprint	yes	yes	yes	yes
Real building height	yes	no	no	no

deliberately focus on attenuation, the primary link-budget quantity and the one for which the dominant propagation structure can be isolated most directly.

The attenuation metric first excludes all building pixels and then retains only finite ray-traced targets in the stored $[20, 185]$ dB support. The target is an unsigned 8-bit array with 1 dB quantization.

Headline metrics use the binary `LineOfSight` field emitted by the configured MATLAB ray tracer and stored as `los_mask`. It is keyed by the city geometry, transmitter position and height, receiver grid, and ray-tracing settings, and is not inferred by the attenuation prior. Building pixels are removed before this mask defines LoS and NLoS receiver sets.

The split is made by city rather than by pixel, tile, or random sample. The official split contains 10840 maps from 37 training cities, 2750 maps from 15 validation cities, and 2590 maps from 14 final test cities. No city is moved between these partitions. All lookup tables, standardization statistics, and regression coefficients use training cities only. Validation is used for development checks; headline results and uncertainty intervals use the final test partition after every fitted quantity has been frozen.

For valid-receiver mask $m_i(p)$, prediction $\hat{y}_i(p)$, and target $y_i(p)$, the pixel-weighted test error is

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i \sum_p m_i(p)[\hat{y}_i(p) - y_i(p)]^2}{\sum_i \sum_p m_i(p)}}. \tag{1}$$

We also report mean absolute error (MAE) and an unweighted macro average of per-city RMSE. RMSE remains the primary pixel-weighted, unit-aware metric. Uncertainty is estimated by 20000 percentile-bootstrap draws over the 14 final test cities. Each draw samples cities with replacement and recomputes the pixel-weighted metric; this preserves within-city spatial dependence rather than treating hundreds of millions of pixels as independent observations.

III. HEIGHT-AWARE ATTENUATION PRIOR

Fig. 1 summarizes calibration and deployment. The method uses known geometry to compute distances, a building/visibility mask to route LoS and NLoS pixels, and local windows over the building-height map. Only compact calibration tables and linear weights are learned. They are frozen before a held-out city is evaluated.

TABLE II
EVALUATION SETTING.

Item	Setting
Map	513×513, 1 m pixels
Carrier	7.125 GHz
Transmitter	Centered; variable height
Receiver	Ground, 1.5 m; buildings masked
Visibility	Configured-ray-tracer LoS output stored in HDF5
Split	10840/2750/2590 train/validation/test maps by city
Test cities	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver

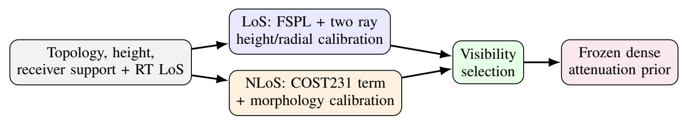

Fig. 1. The prior keeps LoS and NLoS propagation mechanisms separate. All CKM-specific calibration is fitted on training cities and then frozen.

A. LoS Branch

Let d_{2D} be the horizontal transmitter-receiver distance. The direct and image-source reflected distances are

$$\begin{aligned} d_{\mathrm{LoS}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} - h_{\mathrm{rx}})^2}, \\ d_{\mathrm{ref}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} + h_{\mathrm{rx}})^2}. \end{aligned} \tag{2}$$

The free-space term follows the Friis distance dependence, while the coherent correction retains the phase difference between the direct and reflected paths [13]:

$$\mathrm{FSPL} = 32.45 + 20\log_{10}(d_{\mathrm{LoS}}/1000) + 20\log_{10}(f_{\mathrm{MHz}}), \tag{3}$$

$$C_{\mathrm{2ray}} = -20\log_{10}\left|1 + \rho(h_{\mathrm{tx}})\frac{d_{\mathrm{LoS}}}{d_{\mathrm{ref}}} e^{-j[k(d_{\mathrm{ref}} - d_{\mathrm{LoS}}) + \phi(h_{\mathrm{tx}})]}\right|. \tag{4}$$

Here $k = 2\pi f/c$ is the free-space wavenumber, ρ is an effective reflected-field amplitude, and ϕ is a fitted phase offset. The latter two are CKM calibration terms, not material Fresnel coefficients. The deployed LoS prior is

$$\mathrm{PL}_{\mathrm{LoS}} = \mathrm{FSPL} + C_{\mathrm{2ray}} + \beta_{\mathrm{LoS}}(h_{\mathrm{tx}}) + r(d_{2D}, h_{\mathrm{tx}}). \tag{5}$$

Calibration uses valid LoS pixels from calibration-side cities only. In each 5 m height bin, a small grid search selects effective amplitude and phase. A closed-form bias removes the remaining vertical offset. Finally, a smoothed radial table stores the mean residual left by the coherent model. Neighboring height bins are smoothed and linearly interpolated at inference. This structured sequence targets the constructive/destructive rings visible around the centered transmitter without giving the prior enough flexibility to memorize building layouts.

In implementation terms, the LoS correction is a lookup table. The stored entries are the effective ρ, ϕ, and bias curves indexed by transmitter-height bin, together with r indexed by height and radial-distance bin. Inference linearly interpolates neighboring height entries and reads the radial correction for each receiver; it does not refit on the new map. This distinction matters: although the table is data-calibrated, all 14 reported test cities are unseen during table construction. The free-space/two-ray structure supplies extrapolatable geometry, while the lookup table supplies a compact correction learned on other environments.

For each height bin, candidates $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and 48 uniformly spaced phases in $[-\pi, \pi)$ are evaluated. For a candidate pair, the remaining bias is

$$\hat{\beta}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|} \sum_{i \in B_h} \left[y_i - \mathrm{FSPL}_i - C_{2\mathrm{ray},i}(\hat{\rho}, \hat{\phi}) \right]. \tag{6}$$

The pair minimizing the bin RMSE is retained. After smoothing the height curves, the remaining error is averaged by radius and stored as r. This ordering prevents the table from encoding arbitrary two-dimensional building texture.

B. NLoS Branch

The blocked-link branch starts from a COST231/Hata-derived urban trend [14]. Its source domain does not match this 7.125 GHz CKM, so the expression is used as an inspectable basis function rather than a transferred final law. Each map has 513×513 cells at 1 m spacing, with the transmitter projection at cell $(256, 256)$. For a receiver cell $p = (u, v)$, the implementation first computes

$$\begin{aligned} d_{2D}(p) &= 1\,\mathrm{m}\sqrt{(u-256)^2 + (v-256)^2}, \\ d(p) &= \max\{d_{2D}(p), 1\,\mathrm{m}\}, \\ h_t &= \max\{h_{\mathrm{tx}}, 1\,\mathrm{m}\}, \quad h_r = \max\{h_{\mathrm{rx}}, 0.5\,\mathrm{m}\} = 1.5\,\mathrm{m}. \end{aligned} \tag{7}$$

Thus $d_{\mathrm{km}} = \max\{d(p)/1000\,\mathrm{m}, 0.001\}$; all logarithms below receive the numerical value in the stated unit. With $a(h_{\mathrm{rx}}) = (1.1\log_{10} f_{\mathrm{MHz}} - 0.7)h_{\mathrm{rx}} - (1.56\log_{10} f_{\mathrm{MHz}} - 0.8)$, and $\eta(h_t) = 44.9 - 6.55\log_{10} h_t$, define

$$\begin{aligned} \widetilde{\mathrm{PL}}_C &= 46.3 + 33.9\log_{10} f_{\mathrm{MHz}} - 13.82\log_{10} h_t - a(h_r) \\ &\quad + \eta(h_t)\log_{10} d_{\mathrm{km}} + 3. \end{aligned} \tag{8}$$

The +3 dB metropolitan correction is retained inside this empirical basis; Eq. (8) is not claimed to be a directly valid COST231 deployment law at the carrier frequency and heights considered here. The regression input is the clipped map

$$\mathrm{PL}_C = \mathrm{clip}\left(\widetilde{\mathrm{PL}}_C, 0, 185\right). \tag{9}$$

The clipping is applied before PL_C enters the regression. For the elevation-dependent morphology features below, let $\theta(p) = (180/\pi)\tan^{-1}[(h_t - h_r)/d(p)]$.

The visibility mask identifies that a direct path is blocked but not which street canyon, roof edge, or clutter configuration dominates the remaining path. We therefore augment PL_C with exactly defined local quantities. Let $B(p) = \mathbf{1}\{T(p) \neq 0\}$ be the building indicator from topology height T; let $G = 1 - B$; and let $N(p) = \mathbf{1}\{m_{\mathrm{LoS}}(p) \leq 0.5\}G(p)$ be the NLoS-ground indicator. For odd k, define the reflect-padded box mean

$$\mathcal{A}_k[z](p) = \frac{1}{k^2} \sum_{q \in \mathcal{W}_k(p)} z(q). \tag{10}$$

The implemented features are

$$\begin{aligned} \ell_d &= \ln(1 + d_{2D}/1\,\mathrm{m}), & \delta_k &= \mathcal{A}_k[B], \\ h_k &= \mathcal{A}_k[TB]/90\,\mathrm{m}, & n_k &= \mathcal{A}_k[N], \\ \sigma_{\mathrm{sh}} &= 2.3197[\max(90 - \theta, 0)]^{0.2361}, & & \\ \theta_n &= \mathrm{clip}(\theta/90, 0, 1), & k &\in \{15, 41\}. \end{aligned} \tag{11}$$

The shadowing proxy is an empirical elevation feature; its role here is deterministic feature construction, not stochastic shadow-fading generation. Thus, h_k is the window-averaged building height divided by 90 m, including ground pixels as zeros; it is not the mean conditioned only on buildings. The final vector, in implementation order, is

$$\begin{aligned} \mathbf{x} = [&\mathrm{PL}_C, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41}\ell_d, \\ &n_{15}, n_{41}, n_{41}\ell_d, \sigma_{\mathrm{sh}}, \theta_n, n_{41}\theta_n, 1]^\top. \end{aligned} \tag{12}$$

Table III removes any remaining ambiguity by giving the array operation, scale, and interpretation of every regressor. Window sizes are in pixels and therefore metres here; $\mathcal{A}_k$ always includes all k^2 cells after reflect padding, including zeros over ground in the height features. Training standardizes non-intercept features within each fitted regime; coefficients are then transformed back to the raw feature space for the deployed dot product.

The regression regime is observable. From a complete topology map, let D be its building-pixel fraction and H its mean nonzero building height. The class is dense/high-rise if $D \geq 0.2549$ or $H \geq 15.95\,\mathrm{m}$; open/low-rise if $D \leq 0.1957$ and $H \leq 10.91\,\mathrm{m}$; and mixed/mid-rise otherwise. Transmitter-height bins are low for $h_{\mathrm{tx}} \leq 58.12\,\mathrm{m}$, middle up to 103.85 m, and high above it. For each training map, at most 1024 NLoS pixels are selected with a deterministic sample seed (global seed 1234). Let $\mathbf{z}$ contain the standardized non-intercept features and an unpenalized intercept. For regime q, the fitted ridge model is

$$\begin{aligned} \hat{\mathbf{w}}_q &= \arg\min_{\mathbf{w}} \sum_{i \in q} (\mathbf{z}_i^\top \mathbf{w} - y_i)^2 + 10^{-2}\|\mathbf{w}_{\neg b}\|_2^2, \\ \mathrm{PL}_{\mathrm{NLoS}} &= \mathrm{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \end{aligned} \tag{13}$$

In the second line, $\hat{\mathbf{w}}_q$ denotes the algebraically equivalent raw-space coefficients after undoing standardization. All nine topology and height regimes are fitted on the 10840 training maps only. Table IV gives representative coefficients for the middle transmitter-height bin. Their magnitudes must be read together with the feature scales in Table III; they are not

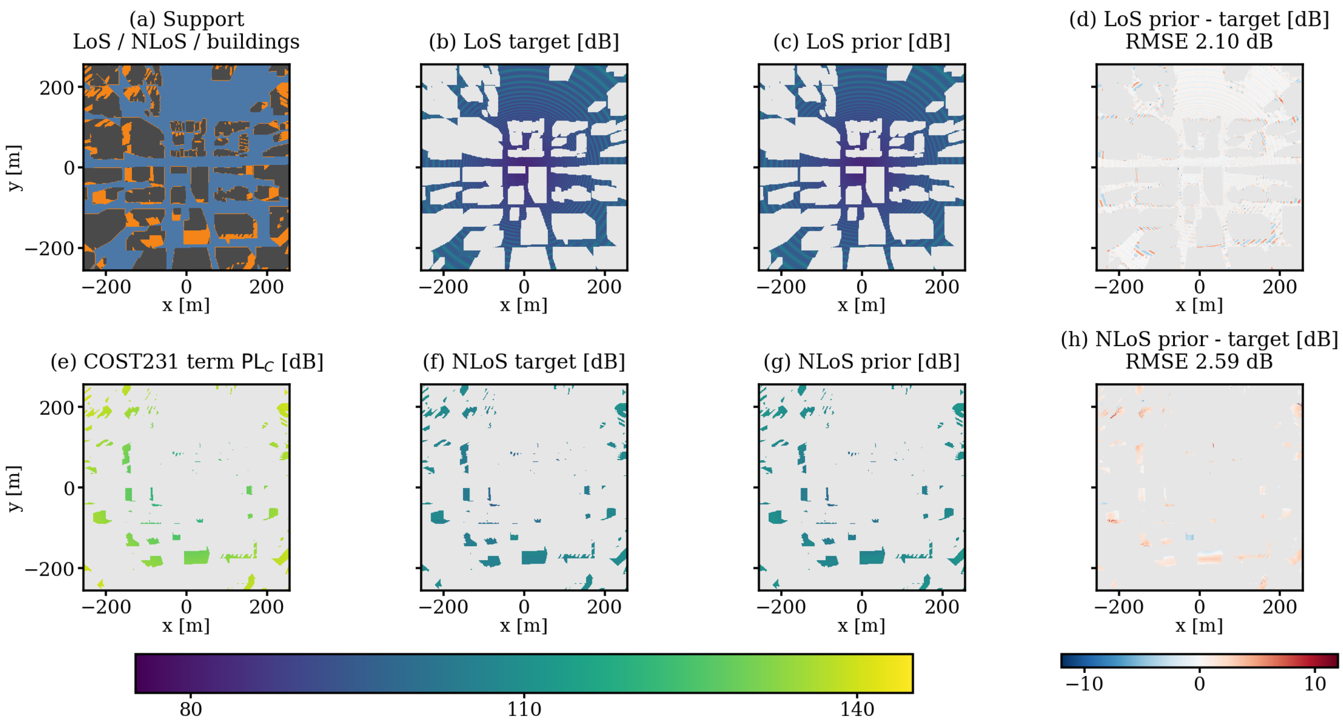

Fig. 2. A held-out example from Vancouver ("sample_15262", $h_{\mathrm{tx}} = 50.30\,\mathrm{m}$, dense/high-rise, low-transmitter regime). Panel (a) shows the geometric support; (b)–(d) are the LoS target, frozen lookup-table prior, and signed error; (e) visualizes the raw COST231 term PL_C; and (f)–(h) give the NLoS target, calibrated prior, and signed error. Grey excludes pixels outside the displayed regime or without a valid target. The unseen map contains 119180 LoS and 25607 NLoS receivers, with 2.10 dB and 2.59 dB RMSE, respectively.

TABLE III

EXACT CONSTRUCTION OF EVERY NLOS MULTILINEAR-REGRESSION INPUT AT RECEIVER CELL p. THE FEATURE ORDER IS ALSO THE COEFFICIENT-VECTOR ORDER. "UNITLESS" MEANS THAT THE STATED METRE OR DEGREE NORMALIZATION HAS ALREADY BEEN APPLIED.

j	Feature $x_j(p)$	Exact computation	Scale and role
1	PL_C	Eq. (9)	dB; clipped COST231 term
2	ℓ_d	$\ln[1 + d_{2D}(p)/1\,\mathrm{m}]$	unitless; radial growth
3	δ_{15}	$15^{-2} \sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; local building fraction
4	δ_{41}	$41^{-2} \sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; contextual building fraction
5	h_{15}	$(15^2\,90\,\mathrm{m})^{-1} \sum_{q\in\mathcal{W}_{15}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; local height mass
6	h_{41}	$(41^2\,90\,\mathrm{m})^{-1} \sum_{q\in\mathcal{W}_{41}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; contextual height mass
7	$\delta_{41}\ell_d$	$\delta_{41}(p)\ell_d(p)$	unitless; density–range interaction
8	n_{15}	$15^{-2} \sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{m_{\mathrm{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; local blocked-ground fraction
9	n_{41}	$41^{-2} \sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{m_{\mathrm{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; contextual blocked-ground fraction
10	$n_{41}\ell_d$	$n_{41}(p)\ell_d(p)$	unitless; blockage–range interaction
11	σ_{sh}	$2.3197[\max\{90 - \theta(p), 0\}]^{0.2361}$	dB; elevation-dependent shadowing proxy
12	θ_n	$\mathrm{clip}[\theta(p)/90, 0, 1]$	unitless; normalized elevation angle
13	$n_{41}\theta_n$	$n_{41}(p)\theta_n(p)$	unitless; blockage–elevation interaction
14	1	an all-ones array with the same 513×513 shape	unitless; intercept

feature importance scores. The accompanying CSV contains all nine 14-term vectors at machine precision.

The topology split changes the fitted relationship, not only its intercept. At middle height, the COST231 coefficient changes sign between open and the other two classes, while the distance, density, local-visibility, and elevation coefficients also change substantially. Correlated raw coefficients cannot be ranked as independent feature importances, but these cross-topology changes show that one global set of slopes would combine distinct propagation regimes.

The current artifact contains all 3×3 NLoS topology/height vectors; the inference code checks compatible height-bin fallbacks only if an exact key is absent. This branch is semi-empirical, but its raw estimate, each feature, selected regime, coefficient, and pre-clipping contribution remain auditable.

The complete prior uses the deterministic LoS mask m_{LoS}:

$$\mathrm{PL}_{\mathrm{prior}} = m_{\mathrm{LoS}}\mathrm{PL}_{\mathrm{LoS}} + (1 - m_{\mathrm{LoS}})\mathrm{PL}_{\mathrm{NLoS}}. \tag{14}$$

Both deployed branches are finally clipped to the target support $[20, 185]$ dB.

TABLE IV
SELECTED RAW-SPACE NLOS COEFFICIENTS FOR THE MIDDLE TRANSMITTER-HEIGHT BIN.

Feature	Open/low-rise	Mixed/mid-rise	Dense/high-rise
PL_C	-0.117224	0.109671	0.117467
ℓ_d	8.669730	4.265489	3.320933
δ_{41}	-14.501030	-29.259829	-33.524686
n_{41}	15.473961	11.327532	7.311767
σ_{sh}	-10.152455	-7.352513	-11.289259
θ_n	-20.718816	-13.965305	-22.991302
Bias	144.446889	119.161245	150.024085

TABLE V
FROZEN PRIOR ON THE 2590-MAP, 14-CITY TEST SET. INTERVALS RESAMPLE CITIES, NOT PIXELS.

Metric	Estimate	95% interval
Overall RMSE [dB]	1.9277	[1.7960, 2.0967]
Overall MAE [dB]	1.2151	[1.1350, 1.3174]
LoS RMSE [dB]	1.7370	[1.6603, 1.8225]
NLoS RMSE [dB]	3.5275	[3.2106, 3.8529]
Macro city RMSE [dB]	1.9823	[1.8484, 2.1293]

TABLE VI
PIXEL-WEIGHTED SIGNED MEAN NLOS REGRESSION CONTRIBUTIONS OVER ALL 31 365 314 VALID NLOS TEST RECEIVERS. ROWS ARE ORDERED BY ABSOLUTE VALUE.

Feature	Mean contribution [dB]
Bias	149.774
σ_{sh}	-72.725
ℓ_d	20.188
PL_C	13.826
δ_{41}	-6.287
$\delta_{41}\ell_d$	4.948
θ_n	-4.454
n_{41}	1.771

TABLE VII
PRIOR RMSE BY HEIGHT ON THE 14-CITY FINAL TEST SET.

Height	Overall	LoS	NLoS	Maps
12–50 m	2.1876	1.7943	3.8133	624
50–150 m	1.9538	1.7788	3.4256	1385
150–300 m	1.6557	1.6370	2.3584	387
300–500 m	1.5591	1.5496	2.2827	194

IV. RESULTS

The component ablation uses the 2750-map validation partition; every fitted variant still uses training cities only. The coherent term reduces LoS RMSE from 3.7900 to 1.7412 dB; the radial table gives a smaller further reduction. The raw COST231 term is not accurate enough by itself. A global NLoS ridge model produces a strong reduction, and observable topology and height regimes improve it further. On final test, topology-only routing retains over 90% of the NLoS RMSE reduction obtained by moving from one global calibration to all nine topology/height regimes; topology is therefore the dominant partition variable, with transmitter-height bins providing a smaller refinement. A topology-specific intercept is already a usable coarse predictor; the heterogeneous slopes supply the remaining gain. A separately recalibrated eight-term variant retaining the 41-pixel context, geometry, visibility, COST231, and bias terms also recovers over 90% of the complete model's improvement over a constant regime baseline.

Table V gives the final attenuation-only test result. The prior reaches 1.9277 dB RMSE and 1.2151 dB MAE across all valid receiver pixels. The LoS error is 1.7370 dB. NLoS remains the hard regime at 3.5275 dB, consistent with the unobserved identity of the dominant diffracted or reflected path after direct-path blockage.

Per-city RMSE ranges from 1.6097 dB in Halifax to 2.5827 dB in Osaka; the unweighted 14-city mean is 1.9823 dB. This spread and the city-bootstrap interval prevent the global score from being read as uniform performance across environments.

After excluding buildings and ground pixels without a valid target, the test metric covers 423 087 119 receivers, of which 7.413% are NLoS. Although the pixel-weighted score is therefore LoS dominated, 424 of the 2590 test maps contain more than 25% NLoS receivers, motivating the separate regime metrics.

Table VI complements the raw coefficients with their actual test-set scale. For each NLoS receiver, its regime-specific coefficient is multiplied by its feature value before clipping; the table reports the signed mean of that term over all valid NLoS receivers. Opposite signs across regimes can cancel in this average, so the values are scale diagnostics rather than causal feature importance scores.

Height is part of the estimator rather than a post-hoc reporting variable. Table VII shows the frozen prior on the final test cities only for four physical height intervals. Error decreases from 2.1876 dB at 12 m to 50 m to 1.5591 dB above 300 m. The largest change is in NLoS, where low transmitters interact more strongly with local rooftops and street canyons. The trend does not imply that altitude alone solves dense prediction; it shows why a single height-independent calibration would combine physically different regimes.

The result is best interpreted as a strong physical/statistical anchor rather than a universal propagation model. The coefficients are tied to the fixed frequency, antenna assumptions, simulator, and available visibility mask. The city split tests transfer across urban layouts, but the labels still come from ray tracing rather than field measurements.

A. Limitations, Visibility, and Runtime

The visibility mask is the configured MATLAB ray tracer's own binary `LineOfSight` output. Identical city geometry, transmitter position and height, receiver grid, and propagation settings therefore define one mask that can be stored and reused. In a repeated five-condition CPU audit, all 742 392 receiver visibility values were identical between runs.

With this stored ray-traced mask, one final prior map takes a median 0.0602 s (95th percentile 0.0844 s) over 100 maps on an AMD Ryzen 5 5600X CPU; HDF5 reading is timed separately at 0.0046 s median. The deployed state contains about 66.8 thousand LoS calibration entries and 126 NLoS regression coefficients. The final NLoS dot product uses at most 14 multiply-accumulate operations per evaluated receiver, although the measured time also includes feature construction and LoS evaluation. For context, an executable PMNet configuration contains 33.34 million parameters and requires 52.71 GMAC for a 256×256 map when counting convolutions only [6]; this is an architectural comparison, not an accuracy comparison across datasets.

On the same CPU, with GPU execution disabled for both paths, MATLAB R2024b ray tracing takes a median 102.289 s (range 91.452 s to 260.310 s) over five Barcelona layouts. Each run uses one transmitter height, a 1 m receiver grid, one reflection, and no diffraction, and extracts attenuation, delay spread, and angular spread. The raw median wall-clock difference is about $1700\times$ for the stored-mask prior core, but it is not a like-for-like speedup because MATLAB returns three channel quantities and also generates visibility. It nevertheless establishes a substantial computational gap at the same spatial resolution. A future journal paper will combine analogous spread priors with distribution-aware deep refinement, enabling a closer multi-output comparison.

All current scenes use flat terrain. Consequently, the reported visibility, distance, and height relationships do not account for terrain-induced blockage or for changes in local ground elevation. Slopes and ridges can alter both the LoS boundary and the meaning of transmitter height relative to nearby receivers. Follow-up work will introduce non-flat digital terrain, compute antenna heights and reflected-path geometry relative to local elevation, add terrain descriptors to the NLoS calibration, and evaluate transfer across relief classes.

V. Conclusion

This paper introduced a dense FR3 UAV CKM dataset with attenuation, delay spread, angular spread, and geometric visibility for 16180 realizations from 66 cities. The modeling contribution deliberately focused on attenuation as a first benchmark. Rather than treating it as generic image-to-image regression, the proposed prior assigns stable structure to explicit components: coherent radial and two-ray behaviour in LoS, COST231 propagation and local morphology in NLoS, and a deterministic visibility mask to route between them. Every lookup table and regression coefficient is fitted on the training cities and frozen before unseen cities are evaluated.

On 2590 maps from 14 held-out cities, the prior reaches 1.9277 dB overall RMSE, 1.7370 dB LoS RMSE, and 3.5275 dB NLoS RMSE. The component study identifies coherent LoS structure and topology-partitioned NLoS calibration as the central ingredients. Topology is the dominant NLoS partition variable: topology-only routing nearly matches the full topology/height calibration, and the fitted slopes change markedly across urban classes. Separate regime reporting remains essential because the aggregate metric is LoS dominated while hundreds of maps contain substantial NLoS support.

The broader conclusion is that a strong CKM estimator should explain stable physical and statistical structure before a more flexible model is asked to learn the remaining local detail. Here that principle yields a compact, auditable CPU estimator with a median runtime of 0.0602 s per map. The prior can therefore serve both as a standalone surrogate and as a frozen anchor for residual learning. Future work will incorporate non-flat terrain, study transfer across frequencies and antenna configurations, and develop a future journal treatment of path loss, delay spread, and angular spread in which deep learning refines the analytical bases and models the remaining output distributions.

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